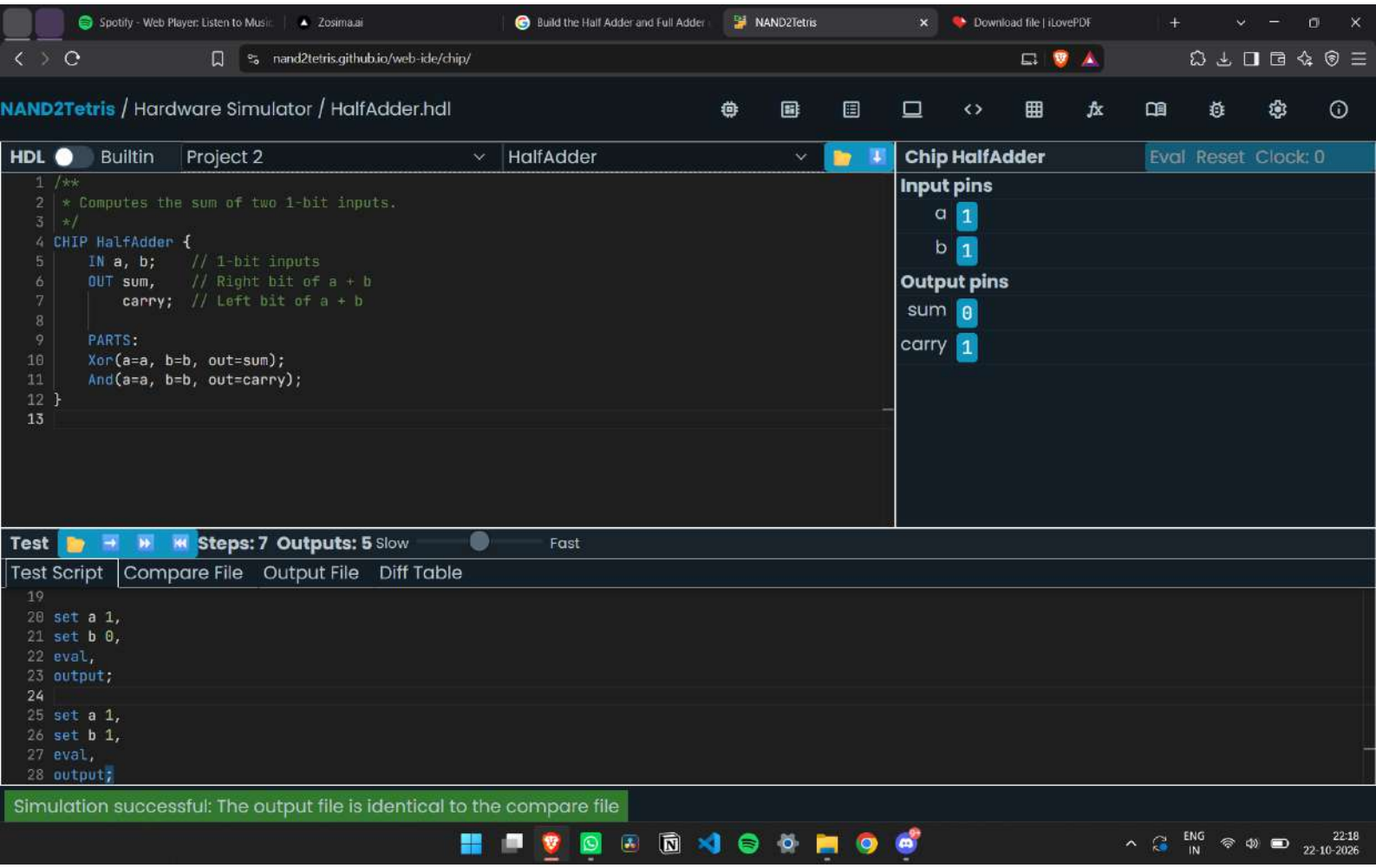
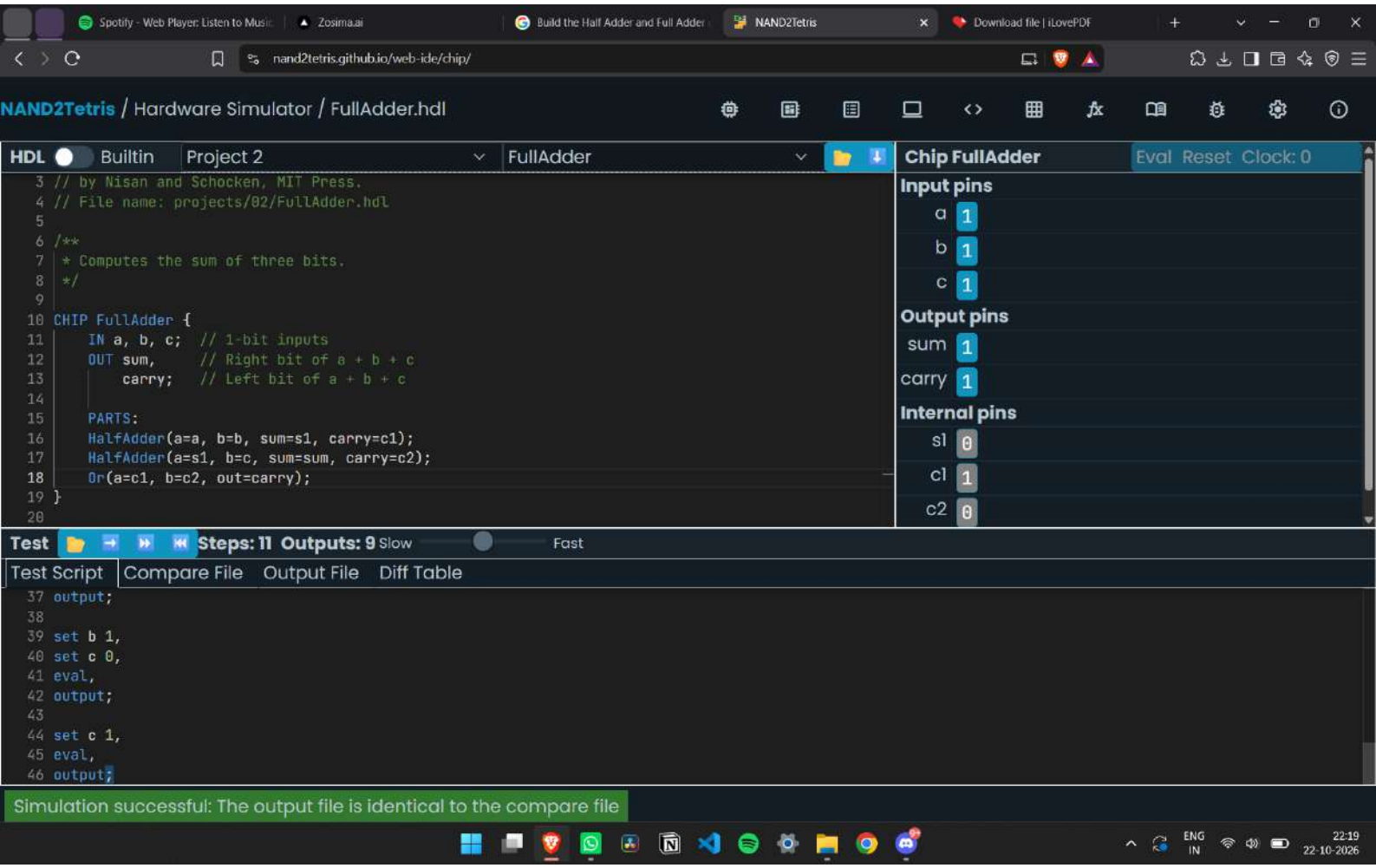

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Build the Half Adder and Full Adder
NAND2Tetris
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NAND2Tetris / Hardware Simulator / HalfAdder.hdl
HDL
Builtin
Project 2
HalfAdder
/**
* Computes the sum of two 1-bit inputs.
*/
CHIP HalfAdder {
IN a, b; // 1-bit inputs
OUT sum, // Right bit of a + b
carry; // Left bit of a + b
PARTS:
Xor(a=a, b=b, out=sum);
And(a=a, b=b, out=carry);
}
Chip HalfAdder
Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
sum 0
carry 1
Test
Steps: 7 Outputs: 5 Slow
Fast
Test Script
Compare File
Output File
Diff Table
set a 1,
set b 0,
eval,
output;
set a 1,
set b 1,
eval,
output;
Simulation successful: The output file is identical to the compare file
ENG
IN
22:18
22-10-2026


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NAND2Tetris / Hardware Simulator / FullAdder.hdl
HDL
Builtin
Project 2
FullAdder
// by Nisan and Schocken, MIT Press.
// File name: projects/02/FullAdder.hdl
/**
* Computes the sum of three bits.
*/
CHIP FullAdder {
IN a, b, c; // 1-bit inputs
OUT sum, // Right bit of a + b + c
carry; // Left bit of a + b + c
PARTS:
HalfAdder(a=a, b=b, sum=s1, carry=c1);
HalfAdder(a=s1, b=c, sum=sum, carry=c2);
Or(a=c1, b=c2, out=carry);
}
Chip FullAdder
Eval Reset Clock: 0
Input pins
a 1
b 1
c 1
Output pins
sum 1
carry 1
Internal pins
s1 0
c1 1
c2 0
Test
Steps: 11 Outputs: 9
Slow
Fast
Test Script
Compare File
Output File
Diff Table
output;
set b 1,
set c 0,
eval,
output;
set c 1,
eval,
output;
Simulation successful: The output file is identical to the compare file
ENG
IN
22:19
22-10-2026